OT and IoT

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Agenda

In this session, we will discuss:

- Introduction to OT
- Components of OT
- OT in Water Treatment Plant
- IoT and its Types
- 7 Levels of IoT
- Protocols of IoT
- Working of IoT

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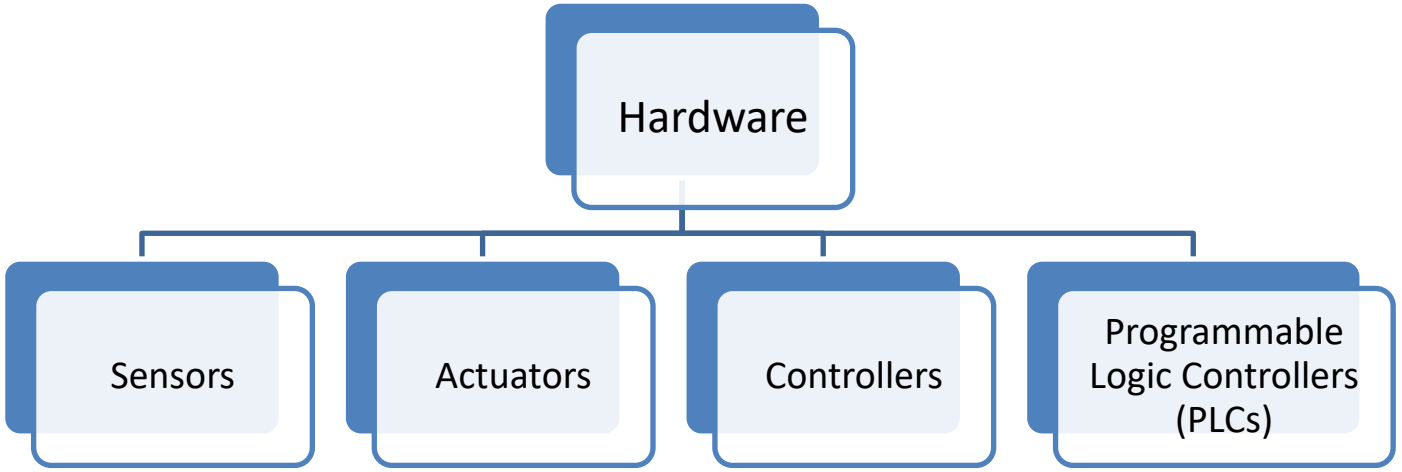
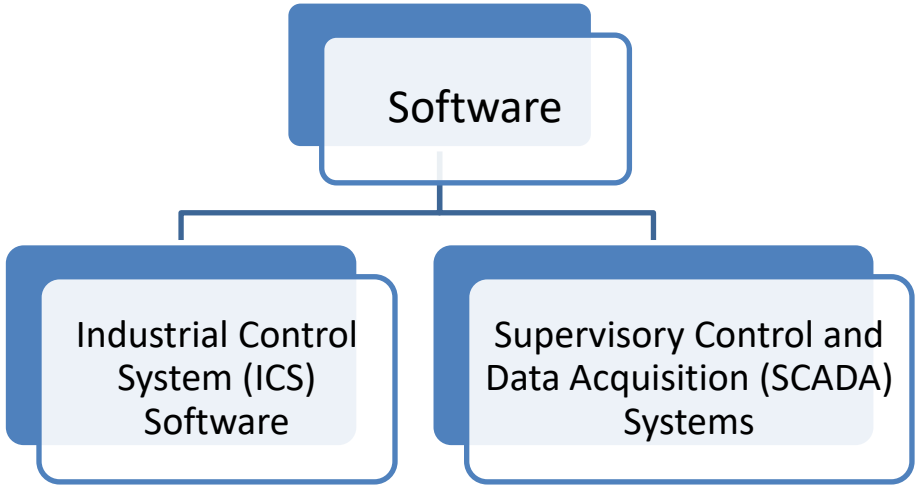
Introduction to Operational Technology (OT)

OT (Operational Technology) is the hardware and software that directly monitors and controls physical devices, processes, and events in the real world.





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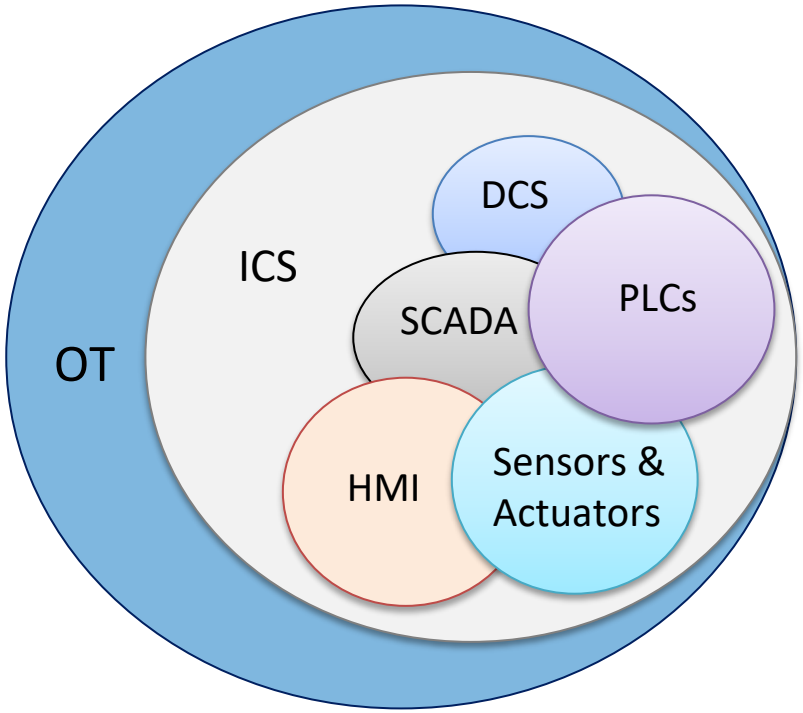
Components of OT

- Industrial Control Systems (ICS)
- Distributed Control Systems (DCS)
- Programmable Logic Controllers (PLCs)
- Supervisory Control and Data Acquisition (SCADA) Systems
- Human-Machine Interface (HMI) Control Units
- Remote Terminal Units (RTUs)
- Communication Links



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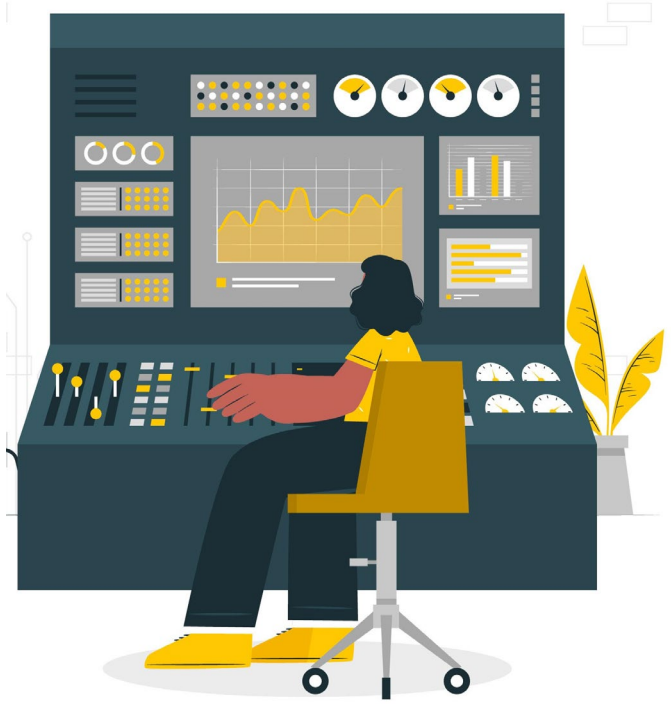
OT in Water Treatment Plant

Sensors - Continuously monitor water quality (e.g., chlorine levels, turbidity).

PLCs - Receive data from the sensors and adjust chemical dosing pumps based on pre-programmed settings.

SCADA - Provides a central interface for operators to monitor plant operations, view real-time data, and adjust settings as needed.



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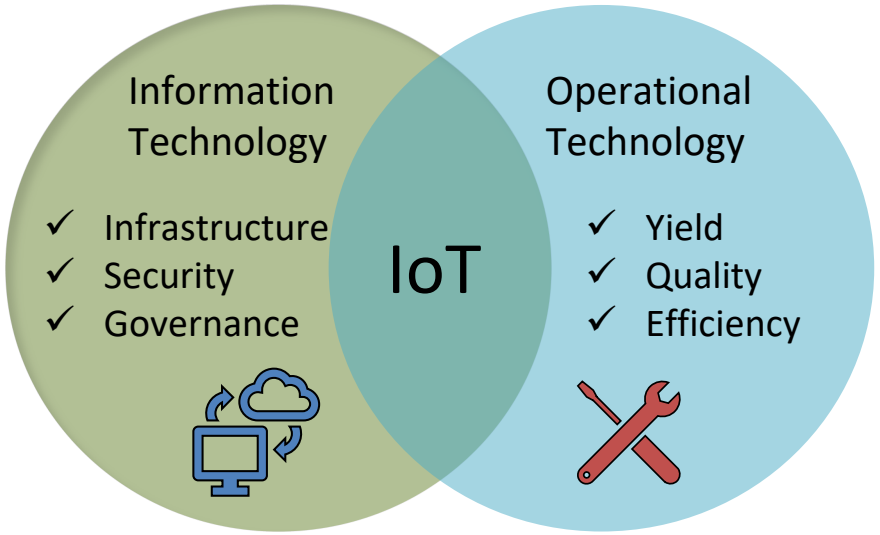
IoT

- IoT (Internet of Things) devices are hardware components that have a wireless network connection and are integrated into equipment or other devices.
- These devices can collect and share data without human intervention.



Types of IoT

- Consumer IoT
- Commercial IoT
- Internet of Military Things (IoMT)
- Industrial Internet of Things (IIoT)
- Infrastructure IoT

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7 Levels of IoT

- Collaboration and Processes (involving people and business processes) Center
- Application (Reporting, Analytics, and Control)
- Data Abstraction (Aggregation and Access)
- Data Accumulation (Storage)
- Edge Computing (Data Element Analysis and Transformation)
- Connectivity (Communication and Processing Units)
- Physical Devices and Controllers

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Protocols of IoT

Application Layer	HTTP MQTT COAP AMQP
Transport Layer	TCP UDP
Network Layer	RPL CORPL CARP 6L0WPAN
Physical and Data Link Layer	Bluetooth ZigBee WIFI

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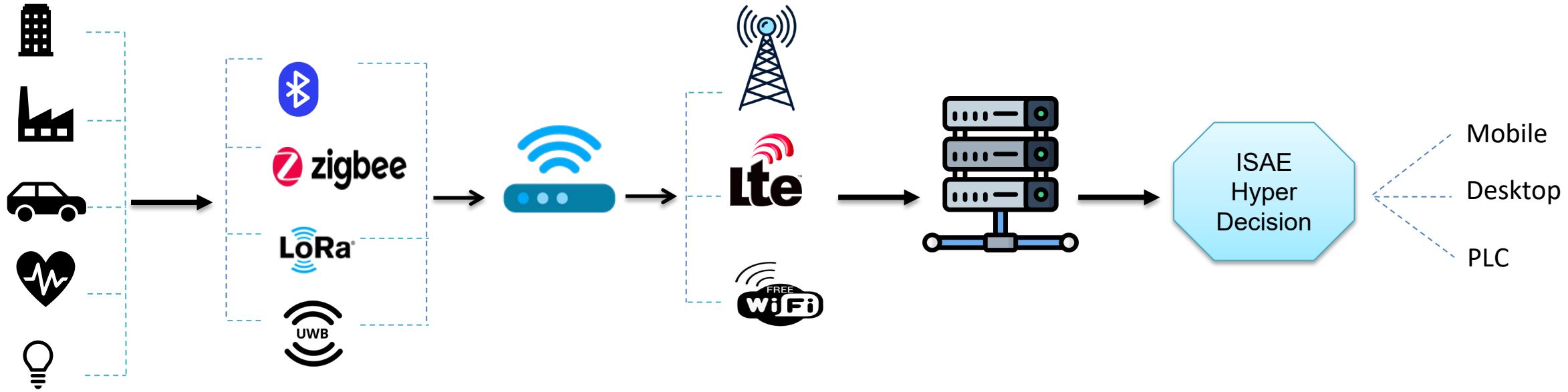
Working of IoT



IoT Sensors >> Wireless Connections >> IoT Gateway >> Carrier >> IoT Server >> Business Rule Engine

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Summary

Here is a brief recap:

- OT focuses on real-time monitoring and control of industrial processes, ensuring reliability and safety in sectors like manufacturing and energy.
- IoT connects everyday objects to the internet, enabling remote monitoring, automation, and data-driven insights across diverse domains such as healthcare, transportation, and smart cities.
- OT collects and processes data within closed systems.
- IoT gathers data from connected devices and transmits it over the internet for centralized analysis.
- OT relies on specialized, often proprietary communication protocols optimized for reliability and low latency.
- IoT uses standardized protocols like HTTP and MQTT for seamless connectivity across devices and platforms.

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